TSM 098

Homework 4

Common factors and introductory factoring

Topics 13-15 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Common divisors and common factors

1. Find the GCD using prime factorizations.

$36, \quad 60$

2. Find the GCD using prime factorizations.

$84, \quad 126$

3. Find the greatest common monomial factor.

$18x^3, \quad 24x^2$

Factor out the GCF

4. Factor out the greatest common factor.

$12x^3 + 18x^2$

5. Factor out the greatest common factor.

$15a^3 - 20a^2$

6. Factor out the greatest common factor.

$8x^2 - 12x + 4$

7. Factor out the greatest common factor.

$14x^4 - 21x^3 + 7x^2$

Guess and check

8. Factor completely over the integers. Check by multiplication.

$x^2 + 7x + 12$

9. Factor completely over the integers. Check by multiplication.

$x^2 - x - 12$

10. Factor completely over the integers. Check by multiplication.

$x^2 - 7x + 10$

11. Factor completely over the integers. Check by multiplication.

$x^2 + 2x - 15$

12. Factor completely over the integers. Check by multiplication.

$x^2 - 25$

13. Factor completely over the integers. Check by multiplication.

$x^2 + 6x + 9$

Choose a first step

14. Factor completely. Start with the GCF.

$3x^2 + 15x + 18$

15. Factor completely. Start with the GCF.

$2x^3 - 18x$

16. Can x squared + 4 be factored into integer linear factors? Explain using possible integer pairs.

17. A student claims x squared + 5x + 6 = (x + 1)(x + 6). Multiply to locate the error, then give the correct factors.